CLOUDLINKNETWORK ENTERPRISE

Network Infrastructure Documentation

v5.0 FINAL • Cisco Packet Tracer 9.0 • March 2026

Company:	**CloudLinkNetwork Enterprise**
Document:	Network Infrastructure Documentation
Version:	**FINAL — All tasks completed**
Prepared by:	NICODEMUS MULILI
Date:	April2026

1. Executive Summary

This document is the final version of the CloudLinkNetwork Enterprise Network Infrastructure Documentation. The network has been fully designed, configured, tested, and verified using Cisco Packet Tracer 9.0. All planned tasks from the original scope have been completed including routing protocols, switching, VLANs, wireless, DHCP, security ACLs, and device management.

> ✓ All 29 implementation tasks are now COMPLETE. The network is fully operational with dual-ISP redundancy, OSPF dynamic routing, HSRP gateway failover, 8-VLAN segmentation, centralized wireless management, and comprehensive security.

Key achievements:

- Core routing — OSPF, HSRP, NAT fully operational across R1, R2, DS1, DS2
- All 7 switches configured — VLANs, trunks, port security, Voice VLAN 70
- DHCP server serving all 8 VLANs via ip helper-address relay
- Security ACLs — Guest isolation, CCTV isolation, HR restriction
- WLC managing all APs — 3 SSIDs with WPA2 encryption
- SSH v2 on all devices — Telnet disabled
- NTP synchronization across all devices
- HSRP and OSPF failover tested and verified

2. Network Topology

2.1 Physical Connections

Device A	Port A	Device B / Port B	Cable Type
R1	Gi0/0	**R2 Gi0/0**	Crossover copper
R1	Gi0/1	**DS1 Gi0/1**	Straight-through copper
R1	Se0/3/0	**ISP Router Se0/1/0**	Serial DCE
R2	Gi0/1	**DS2 Gi0/2**	Straight-through copper
R2	Se0/3/0	**ISP Router Se0/1/1**	Serial DCE
DS1	Gi0/2	**DS2 Gi0/1**	Crossover copper
DS1	Fa0/1	**SW-Branch1-Orange Fa0/1**	Straight-through
DS1	Fa0/2	**SW-Branch2-Red Fa0/1**	Straight-through
DS1	Fa0/3	**SW-Branch3-Green Fa0/1**	Straight-through
DS1	Fa0/4	**SW-Branch4-Yellow Fa0/1**	Straight-through
DS1	Fa0/10	**DHCP Server Fa0**	Straight-through
DS2	Fa0/1	**SW-Branch4-Yellow Fa0/1**	Straight-through
DS2	Fa0/2	**SW-Branch5-Cyan Fa0/2**	Straight-through

2.2 Logical Design

The network uses a three-layer hierarchical design. The Core Layer has R1 and R2 (2901 routers) with dual ISP links, OSPF, HSRP, and NAT. The Distribution Layer uses DS1 and DS2 (3560 L3 switches) for inter-VLAN routing via SVIs and OSPF. The Access Layer has 5 branch switches (2960) serving department zones. The ISP router simulates the internet gate

3. VLAN Design

8 VLANs segment traffic by department and security zone. VLANs 99 and 100 are fully isolated via ACLs. VLAN 70 serves IP phones across Finance, HR/Accounting, and Marketing branches.

VLAN	Name	Network	Gateway	DHCP Range	Department
10	NOC	10.10.10.0/24	10.10.10.1	10.10.10.50-200	NOC — Branch1 Orange
20	CUSTOMERS	10.20.20.0/24	10.20.20.1	10.20.20.50-200	Customers — Branch2 Red
40	HR	10.40.40.0/24	10.40.40.1	10.40.40.50-200	HR — Branch4 Yellow
50	FINANCE-ACCT	10.50.50.0/24	10.50.50.1	10.50.50.50-200	Finance — Branch3 Green
60	MARKETING	10.60.60.0/24	10.60.60.1	10.60.60.50-200	Marketing — Branch5 Cyan
70	VOICE	10.70.70.0/24	10.70.70.1	10.70.70.50-150	IP Phones — Branch3,4,5
99	GUEST-WIFI	172.16.99.0/24	172.16.99.1	172.16.99.10-200	Guest WiFi — All (isolated)
100	CCTV	192.168.100.0/24	192.168.100.1	192.168.100.10-150	CCTV — All (isolated)

4. Device IP Addressing

Device	Interface	IP Address	Subnet Mask	Role
R1	Gi0/1 → DS1	10.0.12.1	255.255.255.252	P2P link to DS1
R1	Se0/3/0	200.10.10.2	255.255.255.252	ISP Link 1
R1	HSRP VIP	192.168.254.254	255.255.255.0	Active gateway (priority 110)
R2	Gi0/1 → DS2	10.0.13.1	255.255.255.252	P2P link to DS2
R2	Se0/3/0	200.20.20.2	255.255.255.252	ISP Link 2
R2	HSRP	192.168.254.254	255.255.255.0	Standby gateway (priority 100)
DS1	Gi0/1 → R1	10.0.12.2	255.255.255.252	Uplink to R1
DS1	Gi0/2 → DS2	10.0.14.2	255.255.255.252	Uplink to DS2
DS1	Vlan10 SVI	10.10.10.1	255.255.255.0	NOC Gateway
DS1	Vlan20 SVI	10.20.20.1	255.255.255.0	Customers Gateway
DS1	Vlan40 SVI	10.40.40.1	255.255.255.0	HR Gateway
DS1	Vlan50 SVI	10.50.50.1	255.255.255.0	Finance Gateway
DS1	Vlan60 SVI	10.60.60.1	255.255.255.0	Marketing Gateway
DS1	Vlan70 SVI	10.70.70.1	255.255.255.0	Voice Gateway
DS1	Vlan99 SVI	172.16.99.1	255.255.255.0	Guest WiFi Gateway
DS1	Vlan100 SVI	192.168.100.1	255.255.255.0	CCTV Gateway
DS2	Gi0/1 → DS1	10.0.14.1	255.255.255.252	Uplink to DS1

Device	Interface	IP Address	Subnet Mask	Role
DS2	Gi0/2 → R2	10.0.13.2	255.255.255.252	Uplink to R2
DS2	Vlan40 SVI	10.40.40.2	255.255.255.0	HR GW (secondary)
DS2	Vlan50 SVI	10.50.50.2	255.255.255.0	Finance GW (secondary)
DS2	Vlan60 SVI	10.60.60.2	255.255.255.0	Marketing GW (secondary)
DS2	Vlan99 SVI	172.16.99.2	255.255.255.0	Guest GW (secondary)
DS2	Vlan100 SVI	192.168.100.2	255.255.255.0	CCTV GW (secondary)
DHCP Server	Fa0 (VLAN10)	10.10.10.10	255.255.255.0	DHCP all VLANs
WLC	Mgmt (VLAN30)	10.30.30.100	255.255.255.0	Wireless controller
ISP Router	Se0/1/0	200.10.10.1	255.255.255.252	ISP Gateway for R1
ISP Router	Se0/1/1	200.20.20.1	255.255.255.252	ISP Gateway for R2

5. Protocols & Technologies

Protocol	Layer	Purpose	Status	Notes
OSPF Area 0	L3	Dynamic routing + failover	✅DONE	R1,R2,DS1,DS2 — all FULL neighbors
HSRP Group 1	L3	Gateway redundancy	✅DONE	R1 Active(110), R2 Standby(100)
VLAN 802.1Q	L2	Traffic segmentation	✅DONE	All 7 switches — 8 VLANs
RSTP	L2	Loop prevention	✅DONE	DS1=root(4096), DS2(8192)
Inter-VLAN SVIs	L3	VLAN routing	✅DONE	DS1 primary, DS2 secondary
NAT/PAT	L3	Internet access	✅DONE	R1 Se0/3/0, R2 Se0/3/0
DHCP Server	App	IP pool management	✅DONE	Server-PT 10.10.10.10 — 8 pools
DHCP Relay	App	Cross-VLAN DHCP	✅DONE	ip helper on DS1+DS2 SVIs
Voice VLAN 70	L2	IP Phone separation	✅DONE	Branch3,4,5 — switchport voice vlan 70
Port Security	L2	MAC-based control	✅DONE	All access switches — sticky MACs
PortFast+BPDU	L2	STP enhancement	✅DONE	All access ports on branch switches
Static Routes	L3	ISP default route	✅DONE	Floating backup on R1 and R2
SSH v2	Security	Encrypted mgmt	✅DONE	All routers and switches
ACL — Guest	Security	Guest WiFi isolation	✅DONE	deny 172.16.99.0 to 10.x.x.x
ACL — CCTV	Security	CCTV isolation	✅DONE	CCTV local-only, no internet
ACL — HR	Security	HR restriction	✅DONE	deny HR VLAN40 to Accounting VLAN50
CAPWAP/WLC	Wireless	AP management	✅DONE	WLC 10.30.30.100, all APs joined
WPA2-PSK	Wireless	WiFi encryption	✅DONE	3 SSIDs — Corp, Guest, Customer
NTP	Management	Time sync	✅DONE	All devices sync to ISP router

6. Complete IOS Configurations

6.1 DHCP Fix — Port Security on Laptop/Device Ports

If a device fails DHCP due to port security violation (err-disabled state), apply this fix on the relevant branch switch:

```
! ===== DHCP FIX — PORT SECURITY ON LAPTOP PORT =====
! If laptop port is err-disabled due to port security:
SW-BranchX# conf t
interface FastEthernet0/X   ! replace X with laptop port
 no switchport port-security violation restrict
 no switchport port-security mac-address sticky
 no switchport port-security
 shutdown
 no shutdown
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 no shutdown
end
```

```
write memory
! Verify:
show port-security interface FastEthernet0/X
```

6.2 SSH v2 — All Devices

Run on every device: R1, R2, ISP Router, DS1, DS2, and all 5 branch switches. This disables Telnet and enables encrypted SSH management.

```
! ===== SSH v2 — ALL ROUTERS AND SWITCHES =====
! Run on: R1, R2, ISP, DS1, DS2, all 5 branch switches
conf t
ip domain-name cloudlinknetwork.local
crypto key generate rsa modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
username admin privilege 15 secret CloudLink@2026!
line vty 0 4
 transport input ssh
 login local
 exec-timeout 5 0
line con 0
 exec-timeout 5 0
 login local
no service telnet
service password-encryption
end
write memory
! Verify:
show ip ssh
show users
```

6.3 Security ACLs — DS1

Three ACLs protect the network: Guest WiFi is limited to internet only, CCTV cameras are fully isolated, and HR cannot access Accounting.

```
! ===== SECURITY ACLs ON DS1 =====
!
! ACL 1: Block HR (VLAN40) from accessing Accounting (VLAN50)
conf t
ip access-list extended RESTRICT-HR
 remark Block HR from Accounting
 deny   ip 10.40.40.0 0.0.0.255 10.50.50.0 0.0.0.255
 permit ip any any
!
! ACL 2: Guest WiFi — internet only, block all internal
ip access-list extended GUEST-RESTRICT
 remark Guest WiFi internet only
 deny   ip 172.16.99.0 0.0.0.255 10.0.0.0 0.255.255.255
 deny   ip 172.16.99.0 0.0.0.255 192.168.0.0 0.0.255.255
 deny   ip 172.16.99.0 0.0.0.255 172.16.0.0 0.0.255.255
 permit ip 172.16.99.0 0.0.0.255 any
!
! ACL 3: CCTV — fully isolated, local traffic only
ip access-list extended CCTV-RESTRICT
 remark CCTV cameras local only
```

```
 permit ip 192.168.100.0 0.0.0.255 192.168.100.0 0.0.0.255
 deny   ip 192.168.100.0 0.0.0.255 any
!
! Apply ACLs to SVIs
interface Vlan40
 ip access-group RESTRICT-HR in
interface Vlan99
 ip access-group GUEST-RESTRICT in
interface Vlan100
 ip access-group CCTV-RESTRICT in
end
write memory
! Verify:
show ip access-lists
show ip interface Vlan40
show ip interface Vlan99
show ip interface Vlan100
```

6.4 Wireless LAN Controller (WLC) + CAPWAP

The WLC at 10.30.30.100 manages all APs via CAPWAP. Three SSIDs are broadcast: Corporate (VLAN40), Guest (VLAN99 isolated), and Customer WiFi (8 zones).

```
! ===== WIRELESS LAN CONTROLLER (WLC) CONFIGURATION =====
! WLC Management IP: 10.30.30.100 (VLAN30)
! Access via browser: http://10.30.30.100 or CLI below
!
! Step 1: Set management interface
config interface address management 10.30.30.100 255.255.255.0 10.30.30.1
config interface vlan management 30
config interface dhcp management primary 10.10.10.10
!
! Step 2: Set AP Manager interface
config interface address ap-manager 10.30.30.101 255.255.255.0 10.30.30.1
!
! Step 3: WLAN 1 — Corporate (VLAN40 HR zone)
config wlan create 1 Corporate-SSID Corp-Net
config wlan interface 1 vlan40
config wlan security wpa enable 1
config wlan security wpa akm psk enable 1
config wlan security wpa akm psk set-key ascii CloudCorp@2026 1
config wlan enable 1
!
! Step 4: WLAN 2 — Guest WiFi (VLAN99, isolated)
config wlan create 2 Guest-SSID CloudGuest
config wlan interface 2 vlan99
config wlan security wpa enable 2
config wlan security wpa akm psk set-key ascii Guest@2026 2
config wlan enable 2
!
! Step 5: WLAN 3 — Customer WiFi (8 zones)
config wlan create 3 Customer-SSID CloudCustomer
config wlan interface 3 management
config wlan security wpa enable 3
config wlan security wpa akm psk set-key ascii Customer@2026 3
config wlan enable 3
!
! Step 6: Verify APs joined WLC
show ap summary
show client summary
show wlan summary
```

6.5 NTP Synchronization

All devices synchronize time to the ISP router for accurate logging and security certificate validation.

```
! ===== NTP SYNCHRONIZATION — ALL DEVICES =====
! Configure ISP router as NTP master
ISP# conf t
ntp master 1
end
!
! Configure all other devices to sync to ISP router
! Run on R1, R2, DS1, DS2, all branch switches:
conf t
ntp server 200.10.10.1
ntp server 200.20.20.1
clock timezone EAT 3   ! East Africa Time (UTC+3)
end
write memory
! Verify:
show ntp status
show clock
```

6.6 HSRP Failover Test

Verify HSRP is working correctly by simulating R1 failure and confirming R2 takes over as the active gateway.

```
! ===== HSRP FAILOVER TEST =====
! Step 1: Verify current state
R1# show standby brief
! Expected: R1 = Active, R2 = Standby
!
! Step 2: Simulate R1 failure
R1# conf t
R1(config)# interface GigabitEthernet0/1
R1(config-if)# shutdown
!
! Step 3: Check R2 takes over
R2# show standby brief
! Expected: R2 = Active within 10 seconds
!
! Step 4: Restore R1
R1(config-if)# no shutdown
R1(config-if)# end
! Expected: R1 reclaims Active role (preempt enabled)
R1# show standby brief
```

6.7 OSPF Reconvergence Test

Verify OSPF reroutes traffic automatically when a link fails.

```
! ===== OSPF RECONVERGENCE TEST =====
! Step 1: Check current routing table
DS1# show ip route
! Step 2: Simulate DS1-R1 link failure
DS1# conf t
DS1(config)# interface GigabitEthernet0/1
DS1(config-if)# shutdown
! Step 3: Verify traffic reroutes via DS2
DS1# show ip route
! Expected: 0.0.0.0/0 now via 10.0.14.1 (DS2 path)
! Step 4: Restore link
```

```
DS1(config-if)# no shutdown
DS1(config-if)# end
! Step 5: Ping test across all paths
DS1# ping 200.10.10.1
DS1# ping 200.20.20.1
DS1# ping 10.40.40.2
```

6.8 End-to-End Ping Verification

These ping tests verify full connectivity, internet access, and ACL enforcement.

```
! ===== END-TO-END PING VERIFICATION =====
! From DS1 — test all VLAN gateways
DS1# ping 10.10.10.10   ! DHCP Server
DS1# ping 10.20.20.1    ! Customers GW
DS1# ping 10.40.40.1    ! HR GW
DS1# ping 10.50.50.1    ! Finance GW
DS1# ping 10.60.60.1    ! Marketing GW
DS1# ping 172.16.99.1   ! Guest WiFi GW
DS1# ping 192.168.100.1 ! CCTV GW
DS1# ping 200.10.10.1   ! ISP Link 1
DS1# ping 200.20.20.1   ! ISP Link 2
!
! From a PC on VLAN40 (HR) — cross-VLAN test
PC> ping 10.50.50.1     ! Should SUCCEED — Finance GW
PC> ping 172.16.99.1    ! Should SUCCEED — Guest GW
PC> ping 8.8.8.8        ! Should SUCCEED — Internet
!
! From Guest WiFi PC (172.16.99.x)
PC> ping 8.8.8.8        ! Should SUCCEED — Internet only
PC> ping 10.40.40.1     ! Should FAIL — ACL blocking
PC> ping 10.10.10.10    ! Should FAIL — ACL blocking
!
! CCTV test
PC> ping 192.168.100.50 ! Should SUCCEED — CCTV to NVR
PC> ping 8.8.8.8        ! Should FAIL — no internet for CCTV
```

6.9 Full Verification Commands

Run these show commands on all devices to confirm every protocol is healthy.

```
! ===== FULL VERIFICATION COMMANDS =====
! Run on DS1:
DS1# show ip ospf neighbor
DS1# show ip route
DS1# show interfaces trunk
DS1# show ip interface brief
DS1# show ip access-lists
!
! Run on DS2:
DS2# show ip ospf neighbor
DS2# show ip route
DS2# show interfaces trunk
DS2# show ip interface brief
!
! Run on R1:
R1# show ip ospf neighbor
R1# show standby brief
R1# show ip nat translations
R1# show ip route
!
! Run on R2:
```

```
R2# show ip ospf neighbor
R2# show standby brief
R2# show ip route
!
! Run on each branch switch:
SW-BranchX# show vlan brief
SW-BranchX# show interfaces trunk
SW-BranchX# show port-security
SW-BranchX# show spanning-tree summary
!
! On WLC:
show ap summary
show client summary
show wlan summary
```

7. DHCP Configuration

The DHCP server at 10.10.10.10 (VLAN10) serves all 8 VLANs via ip helper-address relay on DS1 and DS2 SVIs.

VLAN	Pool Name	Network	Gateway	DHCP Range	Helper
10	**VLAN10-NOC**	10.10.10.0/24	10.10.10.1	10.10.10.50-200	Direct (server on VLAN10)
20	**VLAN20-CUSTOMERS**	10.20.20.0/24	10.20.20.1	10.20.20.50-200	DS1 Vlan20 helper
40	**VLAN40-HR**	10.40.40.0/24	10.40.40.1	10.40.40.50-200	DS1+DS2 Vlan40 helper
50	**VLAN50-FINANCE**	10.50.50.0/24	10.50.50.1	10.50.50.50-200	DS1+DS2 Vlan50 helper
60	**VLAN60-MARKETING**	10.60.60.0/24	10.60.60.1	10.60.60.50-200	DS1+DS2 Vlan60 helper
70	**VLAN70-VOICE**	10.70.70.0/24	10.70.70.1	10.70.70.50-150	DS1+DS2 Vlan70 helper
99	**VLAN99-GUEST**	172.16.99.0/24	172.16.99.1	172.16.99.10-200	DS1+DS2 Vlan99 helper
100	**VLAN100-CCTV**	192.168.100.0/24	192.168.100.1	192.168.100.10-150	DS1+DS2 Vlan100 helper

8. Wireless Network

SSID	Network Name	VLAN	Security	Purpose
Corporate	Corp-Net	VLAN40	WPA2-PSK	Staff devices — HR zone, full internal access
Guest WiFi	CloudGuest	VLAN99	WPA2-PSK	Guests — internet only, fully isolated via ACL
Customer WiFi	CloudCustomer	Management	WPA2-PSK	8 customer zones — dynamic IPs via DHCP

WLC Management IP: 10.30.30.100 (VLAN30). All APs use CAPWAP to discover and join the WLC automatically. Clients roam seamlessly between APs without re-authentication.

9. Security Implementation

Feature	Applied On	Zone	Policy
ACL Guest	DS1 Vlan99	VLAN 99	Internet only — deny all 10.x.x.x, 192.168.x.x, 172.16.x.x internal ranges
ACL CCTV	DS1 Vlan100	VLAN 100	Fully isolated — CCTV to CCTV only, no internet, no internal network access
ACL HR	DS1 Vlan40	VLAN 40	HR cannot access Accounting (VLAN50) — all other VLANs permitted
Port Security	All access SWs	All VLANs	Sticky MAC learning, max 2 MACs per port, violation restrict (CCTV = shutdown)
SSH v2	All devices	Management	RSA 2048-bit, Telnet disabled, exec-timeout 5 min, local auth required
WPA2-PSK	WLC + all APs	Wireless	All 3 SSIDs use WPA2-PSK encryption — no open WiFi networks

10. Budget Estimate (Kenya Shillings)

Prices are approximate 2025-2026 Kenyan market rates (KES). Exchange rate: 1 USD ≈ KES 129. Refurbished Cisco equipment can reduce hardware costs by 30-50%.

#	Item	Qty	Unit (KES)	Total (KES)	Notes
1	Cisco 2901 Router (R1, R2)	2	110,000	**220,000**	Dual WAN, OSPF, HSRP, NAT
2	Cisco 3560-24PS L3 Switch (DS1)	1	420,000	**420,000**	Inter-VLAN routing, OSPF
3	Cisco 3560-24PS L3 Switch (DS2)	1	420,000	**420,000**	Secondary distribution
4	Cisco 2960-24 Access Switch	5	95,000	**475,000**	One per branch zone
5	Cisco WLC 2504 Wireless Controller	1	320,000	**320,000**	CAPWAP, manages all APs
6	Cisco Aironet 1702i AP (indoor)	8	78,000	**624,000**	Branch + main APs
7	Cisco ISP Router (3rd router)	1	110,000	**110,000**	Simulates ISP gateway
8	CCTV IP Cameras PoE 1080p	12	22,000	**264,000**	VLAN100 isolated
9	NVR Network Video Recorder	1	90,000	**90,000**	CCTV storage, VLAN100
10	Cisco IP Phones 7841 series	10	35,000	**350,000**	VLAN70 Voice
11	Server-PT DHCP/DNS Server	1	85,000	**85,000**	HP ProLiant or equiv.
12	Cisco ASA 5506-X Firewall	1	230,000	**230,000**	Perimeter security
13	UPS APC Smart 1500VA	4	45,000	**180,000**	Power protection
14	Cat6 Cable 305m box	4	15,000	**60,000**	Structured cabling
15	Patch Panel 24-port	4	10,000	**40,000**	Cable management
16	19-inch Server Rack 24U	2	52,000	**104,000**	Equipment housing
17	SFP Modules fiber uplinks	8	11,500	**92,000**	Gi uplinks
18	PRTG Network Monitor 1yr	1	155,000	**155,000**	Monitoring & alerting
19	Wireless Routers per branch	5	12,000	**60,000**	Dept + Guest WiFi
20	Professional Installation & Config	1	380,000	**380,000**	Cabling + IOS + testing

#	Item	Qty	Unit (KES)	Total (KES)	Notes
21	Contingency 10%	—	—	**469,300**	Buffer for unforeseen costs
	TOTAL ESTIMATED BUDGET			**KES 5,162,300**	**≈ USD 40,000**

Cost Category	Sub-Total (KES)
Routing & Switching Hardware	KES 1,725,000
Wireless Infrastructure	KES 1,004,000
CCTV System	KES 354,000
IP Phones (Voice VLAN)	KES 350,000
Servers & Security	KES 315,000
Passive Infrastructure	KES 296,000
Software & Licensing	KES 155,000
Labor & Contingency	KES 849,300
TOTAL	**KES 5,162,300**

11. Implementation Checklist

All 29 tasks completed. Full implementation verified via show commands and live ping tests.

#	Task	Status	Notes
1	DS1 ip routing + SDM template	✅DONE	show ip route verified
2	VLANs on all 7 switches	✅DONE	show vlan brief confirmed
3	DS1 SVIs all 8 VLANs	✅DONE	All up/up after trunks
4	DS2 SVIs VLANs 40,50,60,99,100	✅DONE	Verified
5	OSPF R1,R2,DS1,DS2 FULL	✅DONE	show ip ospf neighbor
6	Dual default routes R1+R2	✅DONE	O*E2 0.0.0.0/0 via both
7	ISP Router serial links	✅DONE	200.10/200.20 in routing table
8	DS1 trunk Fa0/1-4 to branches	✅DONE	show interfaces trunk
9	DS2 trunk Fa0/1-2 to Branch4+5	✅DONE	CDP confirmed
10	SW-Branch1-Orange (NOC)	✅DONE	VLANs + trunks verified
11	SW-Branch2-Red (Customers)	✅DONE	VLANs + trunks + CCTV
12	SW-Branch3-Green (Finance+Voice)	✅DONE	VLANs + voice VLAN70
13	SW-Branch4-Yellow (HR+Accounting)	✅DONE	VLANs + voice VLAN70
14	SW-Branch5-Cyan (Marketing+Voice)	✅DONE	VLANs + voice VLAN70
15	DHCP Server 8 pools	✅DONE	Server-PT configured
16	DHCP Relay DS1+DS2	✅DONE	ip helper all SVIs
17	DHCP all branches tested	✅DONE	PCs getting IPs
18	Port security sticky MACs	✅DONE	All branch switches
19	SSH v2 all devices	✅DONE	Telnet disabled

#	Task	Status	Notes
20	ACL Guest WiFi isolation	✅**DONE**	172.16.99.0 internet only
21	ACL CCTV isolation	✅**DONE**	192.168.100.0 local only
22	ACL HR restriction	✅**DONE**	VLAN40 blocked from VLAN50
23	WLC + CAPWAP APs	✅**DONE**	All APs joined WLC
24	3 SSIDs configured	✅**DONE**	Corp, Guest, Customer
25	HSRP failover test	✅**DONE**	R2 took over correctly
26	OSPF reconvergence test	✅**DONE**	Traffic rerouted in <5 sec
27	End-to-end ping all VLANs	✅**DONE**	All 100% success
28	NTP synchronization	✅**DONE**	All devices synced
29	Final documentation	✅**DONE**	v5.0 Final

12. Revision History

Status:	**IMPLEMENTATION COMPLETE ✓**
Budget (KES):	**KES 5,162,300 (≈ USD 40,000)**

Ver	Date	Changes	Author
1.0	Feb 2026	Initial topology design	Network Engineering
2.0	Mar 2026	Updated from DS2 show vlan brief, VLAN99 gap identified	Network Engineering
3.0	Mar 2026	Full switch configs DS1+DS2+5 branches	Network Engineering
4.0	Mar 2026	OSPF verified, DHCP configured, Voice VLAN70, budget in KES	Network Engineering
5.0	APRIL 2026	FINAL — All tasks complete: SSH, ACLs, WLC, NTP, tests	Network Engineering